Mandelstam Variables: Complete Reference

General Theory + Compton Scattering Application

PHYS 7502

1 General Definition

Mandelstam Variables for $2 \to 2$ Scattering

For any process $A(p_A)+B(p_B) \to C(p_C)+D(p_D)$, the three Lorentz-invariant Mandelstam variables are:

$$s \equiv (p_A + p_B)^2 = (p_C + p_D)^2 \tag{1}$$

$$t \equiv (p_A - p_C)^2 = (p_B - p_D)^2 \tag{2}$$

$$u \equiv (p_A - p_D)^2 = (p_B - p_C)^2 \tag{3}$$

Physical meaning:

- s = (total 4-momentum)^2 = square of the center-of-mass energy
- t = (momentum transfer from A to C)^2
- u = (momentum transfer from A to D)^2

2 The Constraint

Sum Rule

$$\boxed{s + t + u = m_A^2 + m_B^2 + m_C^2 + m_D^2} \tag{4}$$

Proof:

$$\begin{aligned} s + t + u &= (p_A + p_B)^2 + (p_A - p_C)^2 + (p_A - p_D)^2 \\ &= p_A^2 + 2p_A{\cdot}p_B + p_B^2 + p_A^2 - 2p_A{\cdot}p_C + p_C^2 + p_A^2 - 2p_A{\cdot}p_D + p_D^2 \\ &= 3p_A^2 + p_B^2 + p_C^2 + p_D^2 + 2p_A{\cdot}(p_B - p_C - p_D) \\ &= 3p_A^2 + p_B^2 + p_C^2 + p_D^2 + 2p_A{\cdot}(-p_A) \\ &= 3m_A^2 + m_B^2 + m_C^2 + m_D^2 - 2m_A^2 \\ &= m_A^2 + m_B^2 + m_C^2 + m_D^2 \end{aligned} \tag{5}$$

where we used 4-momentum conservation $p_B - p_C - p_D = -p_A$.
Only **two** of the three variables are independent (the third is fixed by the sum rule plus the masses).

3 Expanding in Terms of Dot Products

Mandelstam Variables as Dot Products

$$s = m_A^2 + m_B^2 + 2\,p_A{\cdot}p_B = m_C^2 + m_D^2 + 2\,p_C{\cdot}p_D \tag{6}$$

$$t = m_A^2 + m_C^2 - 2\,p_A{\cdot}p_C = m_B^2 + m_D^2 - 2\,p_B{\cdot}p_D \tag{7}$$

$$u = m_A^2 + m_D^2 - 2\,p_A{\cdot}p_D = m_B^2 + m_C^2 - 2\,p_B{\cdot}p_C \tag{8}$$

Solving for the dot products:

$$p_A{\cdot}p_B = \tfrac{1}{2}(s - m_A^2 - m_B^2) \tag{9}$$

$$p_A{\cdot}p_C = \tfrac{1}{2}(m_A^2 + m_C^2 - t) \tag{10}$$

$$p_A{\cdot}p_D = \tfrac{1}{2}(m_A^2 + m_D^2 - u) \tag{11}$$

4 Physical Region

Which Variable is Positive?

For a physical scattering process $A + B \to C + D$:

- $s > 0$ (total CM energy squared is always positive)
- $t \leq 0$ (spacelike momentum transfer, unless threshold effects)
- $u \leq 0$ (spacelike momentum transfer)

The name of each "channel" tells you which variable is timelike (> 0):

- **s-channel:** $s > 0$. Direct scattering or annihilation. Intermediate particle has invariant mass $\sqrt{s}$.
- **t-channel:** $t > 0$. Crossed process. The "same" amplitude evaluated in a region where t is the CM energy.
- **u-channel:** $u > 0$. Another crossed process.

5 Center-of-Mass Frame Expressions

CM Frame

In the CM frame, $\vec{p}_A + \vec{p}_B = 0$, with initial momentum magnitude $|\vec{p}|$ and final momentum magnitude $|\vec{p}'|$, scattering angle θ^*:

$$s = (E_A^* + E_B^*)^2 = E_{\text{cm}}^2 \tag{12}$$

$$t = m_A^2 + m_C^2 - 2E_A^* E_C^* + 2|\vec{p}||\vec{p}'|\cos\theta^* \tag{13}$$

$$u = m_A^2 + m_D^2 - 2E_A^* E_D^* - 2|\vec{p}||\vec{p}'|\cos\theta^* \tag{14}$$

For **massless particles** ($m_A = m_B = m_C = m_D = 0$):

$$s = 4E^{*2} \tag{15}$$

$$t = -2E^{*2}(1 - \cos\theta^*) = -\frac{s}{2}(1 - \cos\theta^*) \tag{16}$$

$$u = -2E^{*2}(1 + \cos\theta^*) = -\frac{s}{2}(1 + \cos\theta^*) \tag{17}$$

where $E^* = \sqrt{s}/2$ is the energy of each particle in the CM frame.

6 Massless Limit Simplifications

All Masses Zero: $m_A = m_B = m_C = m_D = 0$

$$\boxed{s + t + u = 0} \tag{18}$$

All dot products simplify to:

$$p_A \cdot p_B = s/2 \qquad p_A \cdot p_C = -t/2 \qquad p_A \cdot p_D = -u/2 \tag{19}$$

$$p_C \cdot p_D = s/2 \qquad p_B \cdot p_D = -t/2 \qquad p_B \cdot p_C = -u/2 \tag{20}$$

Useful angular relations:

$$-\frac{t}{s} = \frac{1 - \cos\theta^*}{2} = \sin^2\frac{\theta^*}{2} \tag{21}$$

$$-\frac{u}{s} = \frac{1 + \cos\theta^*}{2} = \cos^2\frac{\theta^*}{2} \tag{22}$$

$$\frac{t}{u} = \frac{1 - \cos\theta^*}{1 + \cos\theta^*} = \tan^2\frac{\theta^*}{2} \tag{23}$$

7 Application: Compton Scattering

Compton in the Professor's Convention

$e^-(p_1) + \gamma(p_2) \to \gamma(p_3) + e^-(p_4)$
$(1 = e^- \text{ in}, 2 = \gamma \text{ in}, 3 = \gamma \text{ out}, 4 = e^- \text{ out})$

$$s = (p_1 + p_2)^2 \qquad s\text{-channel propagator: virtual } e^- \text{ with } q = p_1 + p_2 \tag{24}$$

$$t = (p_1 - p_3)^2 \qquad t\text{-channel propagator: virtual } e^- \text{ with } q = p_1 - p_3 \tag{25}$$

$$u = (p_1 - p_4)^2 \qquad (\text{no } u\text{-channel diagram at tree level}) \tag{26}$$

On-shell: $p_1^2 = p_4^2 = m^2,\ p_2^2 = p_3^2 = 0$.
Sum rule: $s + t + u = 2m^2$.
At $m = 0$: $s + t + u = 0$, and:

$$p_1{\cdot}p_2 = s/2 \qquad p_1{\cdot}p_3 = -t/2 \qquad p_1{\cdot}p_4 = -u/2 \tag{27}$$

$$p_3{\cdot}p_4 = s/2 \qquad p_2{\cdot}p_4 = -t/2 \qquad p_2{\cdot}p_3 = -u/2 \tag{28}$$

8 Application: Pair Annihilation

Pair Annihilation

$e^-(q_1) + e^+(q_2) \to \gamma(q_3) + \gamma(q_4)$

$$s = (q_1 + q_2)^2 \qquad \text{CM energy of } e^+e^- \text{ pair} \tag{29}$$

$$t = (q_1 - q_3)^2 \qquad t\text{-channel propagator: virtual } e^- \text{ with } q = q_1 - q_3 \tag{30}$$

$$u = (q_1 - q_4)^2 \qquad u\text{-channel propagator: virtual } e^- \text{ with } q = q_1 - q_4 \tag{31}$$

On-shell: $q_1^2 = q_2^2 = m^2,\ q_3^2 = q_4^2 = 0$.
Sum rule: $s + t + u = 2m^2$.
Crossing from Compton: $s_C \to t_A,\ t_C \to u_A,\ u_C \to s_A$.

9 Quick Reference Table

Mandelstam Variables at a Glance ($m = 0$)

Quantity	Formula	CM Frame
s	$(p_A + p_B)^2$	E_{cm}^2
t	$(p_A - p_C)^2$	$-\frac{s}{2}(1 - \cos\theta^*)$
u	$(p_A - p_D)^2$	$-\frac{s}{2}(1 + \cos\theta^*)$
$s + t + u$	$\sum m_i^2$	0 (if all massless)
$-t/s$		$\sin^2(\theta^*/2)$
$-u/s$		$\cos^2(\theta^*/2)$